



Larval development, juvenile survival, and burrowing rate of geoduck clams (*Panopea japonica*) under different pH conditions

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Abstract

Changes in seawater pH in the culture environment have numerous effects on marine bivalves. To investigate the effect of pH on larval development, juvenile survival, and burrowing rate of geoduck clams (*Panopea japonica*), specimens were cultured under different experimental pH conditions (6.8, 7.2, 7.6, 8.0, 8.4, and 9.2). The pH range for optimal growth and development differed among different larval stages. However, significant changes in larval development ($p < 0.5$) relative to the control (pH 8.0) were detected at very high or low pH. The best hatching rate occurred at pH 7.6–8.8, and it was significantly lower ($p < 0.5$) at pH 6.8, 7.2, and 9.2. Larval survival was only 4% at pH 6.8. Growth rate of larvae increased with increasing pH and the optimal range was 8.0 to 8.8. This finding suggests that alkaline conditions favor growth and development of larvae and allow them to reach the metamorphic stage promptly. Prolonged metamorphosis was a common feature in larvae in the different experimental pH groups. pH below 7.6 and above 8.8 resulted in a significantly ($p < 0.5$) lower metamorphosis rate, suggesting that high acidity and alkalinity were deleterious to larval metamorphosis. Thus, the suitable pH range for metamorphosis was 8.0 to 8.8. Lower and higher pH also affected the digging behavior of geoduck clams, and the suitable pH range for burrowing was 7.2 to 8.8. Geoduck clams stopped burrowing in the sand at pH 6.8 and 9.2, possibly due to impairment of biological process caused by acidic and alkaline seawater. These findings provide valuable information about the larval and juvenile responses of *P. japonica* in variable pH.

Keywords pH · Growth · Development · Suitable condition · Hatching · Burrowing · Metamorphosis · *Panopea japonica*

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Introduction

The geoduck clam is the largest burrowing clam in the world. Average annual production of geoduck is around 6000 metric tons (Shamshak and King 2015). *Panopea japonica* is distributed mainly in northern China, Japan, and Korea (Rocha-Olivares et al. 2010), where it is in high demand. The edible portion of *P. japonica* is 50–55% having high in protein and fatty acid value (You et al. 1993). This nutritional value prompted farmers to culture geoduck clams in China for human consumption. In recent years, aquaculture of *P. japonica* has drawn significant attention, due to its demand and profitability (Huo et al. 2017). However, variations in environmental conditions that deleteriously impact larval growth and survival.

pH is considered as one of the main factors affecting water quality. It plays a central role in determining the environmental setting and the geochemistry of oceans (Mackenzie and Garrels 1996). The chemical processes that reduce the pH of seawater have come to be known as ocean acidification (Caldiera and Wickett 2003; Feely et al. 2004; Orr et al. 2005). Atmospheric and anthropogenic CO₂ (combustion of fossil fuels) influx cause fluctuation of seawater pH (Thacker et al. 2005; Hudson et al. 2007; Ivanina and Sokolova 2015), which influence water quality and eventually marine bivalves. In closed culture system, variation in pH occur as acid is produced during the nitrification process and as a consequence of CO₂ being released through respiration of the organisms (Bennich 2015).

Over the last decades, an increasing attention has been paid to the potential impacts of pH fluctuation on marine organism (Scanes et al. 2014; Shi et al. 2016; Xu et al. 2016; Zhao et al. 2016). Though physiological processes operate normally in some aquatic biota under a relatively wide pH range, normal physiological and biological functions of bivalves found to be vulnerable to the variable pH (Kurihara et al. 2004). In addition, sensitivity of marine bivalves to extreme pH conditions varies with the species and age (Kroeker et al. 2010). Bivalves and other invertebrate are in most cases show lower tolerance at early life stages (Parker et al. 2010; Van Colen et al. 2012). Gazeau et al. (2010) reported to have a reduced D-larval growth and survival rate in *Mytilus edulis* at low pH in seawater. Juvenile giant clam, *Tridacna squamosa* also show similar effect encounterend with varied ocean pH (Watson 2015).

The impact of fluctuating pH levels on clam are expected to be dramatic. A change in pH can alter gene expression patterns, cause morphological changes, lead to reduced calcification and growth rates, and ultimately result in mortality (Green et al. 2004; Zippay and Hoffman 2010). Bivalve larval development goes through several stages. Fertilized embryos initially develop into non-feeding trochophore larvae, then into feeding veliger larvae, and finally into post-larvae. All of these stages are influenced by changes in pH of the culture environment (Jablonski and Lutz 1982). Several studies reported negative impacts of pH change on early life stages of marine bivalves such as *Crassostrea virginica* and *Mercenaria mercenaria* (Michaelidis et al. 2005), *Saccostrea glomerata* (Wilson and Hyne 1997), *Sphaerechinus granularis* (Cipollaro et al. 1986), *Tivela stultorum* (Alvarado-Alvarez et al. 1996), and *Placopecten magellanicus* (Desrosiers et al. 1996). These deleterious effects mainly occur in the larval development stages.

To date, study on geoduck documented sensitivity to environmental stressors, including temperature, salinity, diet, and stocking density (Huo et al. 2017), however, there are no studies investigating the impact of pH. The goal of this study was to evaluate the effect future ocean pH levels on the development of early larval stages, juvenile survival, and burrowing rate of geoduck clams in commercial farm.

Materials and methods

Brood stock

This study was conducted in the hatchery of Zhangzi Island Limited Corporation of Dalian, Liaoning Province, China, in June–September 2015. Sexually mature geoduck clam individuals with a shell length of 10.67 ± 1.08 cm were collected from the northern part of the Yellow Sea. The clams were cultured indoors at 8–13 °C for 2 weeks in 25 m³ concrete tanks. Fertilized eggs were collected using a 30- μ m nylon sieve when the parental individuals naturally spawned. These eggs were transported to Key Laboratory of Dalian Ocean University for use in the experiments.

Experiment I: Larval development under different pH conditions

pH levels were set at 6.8, 7.2, 7.6, 8.0, 8.4, and 9.2. The pH 8.0 group was used as the natural seawater control group. pH was adjusted with 40 g/L sodium hydroxide solution and 1:10 hydrochloric acid. This method was used to adjust the pH to the predetermined value in the test tank, keeping the error within ± 0.05 . Water was changed once a day, and total ammonia, nitrite, and nitrogen content, pH, water temperature, and salinity were measured before and after changing the water. Exchange of carbon dioxide between the water and the air was prevented to maintain a relatively stable pH during the test.

After approximately 72 h, the fertilized eggs described in the previous section developed into straight-hinged larvae (D-larvae). The D-larvae were collected using a 50- μ m nylon sieve and transferred into 60-L tanks for larval rearing in seawater at the corresponding pH. Three replicates were used per pH treatment. D-larva was placed in a 60-L container with an initial density of 8–10 larvae per mL. Each day during larval rearing, the water was completely exchanged with sand-filtered seawater of the corresponding pH, and the salinity, temperature, and total ammonia and nitrite concentrations were maintained at 32 ± 0.5 ppt, 19 ± 1 °C, 0.169 ± 0.012 mg/L, and 0.007 ± 0.003 mg/L, respectively.

The larvae were fed *Isochrysis galbana* during days 1–3 and a mixture of *I. galbana* and *Nitzschia closterium* (1:1) from day 4 to the juvenile stage. The feeding ration was increased as the larval development progressed; the larvae were fed a ration of 2000–4000 cells/mL/day on days 1–3 and 4000–8000 cells/mL/day from day 4 until metamorphosis.

Maintaining the density of each group of larvae was accomplished by adjusting the water volume every 2 days. The larval density was reduced during larval growth from the initial 8–10 larvae per mL to 1–2 larvae after metamorphosis. Parameters measured during the experiment are described in the “[Measurements](#)” section

Experiment II: Juvenile survival and burrowing rate under different pH conditions

The same pH levels were used in this experiment. Thirty healthy juveniles of similar size (0.78 ± 0.05 mm) were placed in seawater and held for 24 h, during which time they were not fed. Test water volume was 3 L, and sand was placed on the bottom of the tanks as the substrate. The sand was sterilized by heating to 100 °C for 90 min followed by fresh water immersion for 24 h prior to use in the tank. The sand was placed in the 3-L tanks. Individuals that were positioned in the tanks could be observed easily were considered to estimate burrowing rates. Burrowing rates were assessed by the 24-h ratio of the number of clams which were

completely buried in the sediment versus the number of those whose valves remained above the surface of the sediment.

Rearing density was one individual per container square centimeter (about 20,000/m²), and the temperature was maintained at 19 °C. During the nursery stage (3 days), 50% of the water was exchanged daily with sand-filtered seawater of the corresponding pH. The juveniles were fed a mixture of *I. galbana* and *N. closterium* (1:1) at a ration of 10,000 cells/mL/day. The experiment was replicated three times using different sets of tanks. Parameters measured during the experiment are described in the “Measurements” section.

Measurements

All tests were based on the measured average of pH as the true value for each pH. Determination of water chemical indicators was strictly implemented in accordance with the ‘Marine Monitoring Code’. Salinity, water temperature, dissolved oxygen content, and pH were measured using an YSI QUATRO multifunctional water quality analyzer (Fisher Scientific, New Hampshire, U.S.A). Total ammonia and nitrogen concentrations were determined by sodium bromide oxidation and the diazo-azo colorimetric method (Weifang Qiangyuan Chemical Industry Co., Liaoning, China), respectively.

For experiment I, 2 ml water samples from each three replicate were taken, and the numbers of eggs and live larvae were counted under a microscope before and after hatching respectively, to estimate density. The hatching rate was calculated based on the production of D-larvae from fertilized eggs. The larval survival rate was calculated based on the total number of live larvae on the day 16 after the D-larval stage. A total of 30 randomly selected larvae were preserved with 4% formaldehyde, and the shell length of each larva was measured under a microscope on day 16 of the larval stage. The metamorphosis rate was calculated by the ratio of the metamorphosed juveniles/pediveliger larvae (when the velum had disappeared and the subsequent shell became obvious).

For experiment II, the juvenile survival rate was calculated as the ratio of the number of normal juveniles: initial number of juveniles. It was recorded on day 3 of the juvenile experimental stage. The number of juveniles burrowed in the latent sand after 3 days was recorded.

Data analysis

Data were transformed (log or arcsine) as necessary to meet the assumptions of the statistical methods employed. The effects of the different pH conditions on larval development, juvenile survival, and burrowing rate of geoduck clams were analyzed with one-way analysis of variance (ANOVA) and Tukey’s multiple comparison tests as implemented in the statistics program R.

Results

Water chemistry

Table 1 shows the pH of seawater within each treatment. The desired pH levels were successfully reached within each treatment.

Table 1 Seawater pH (mean $\pm$ SD) during the experimental period

Experimental group	A	B	C	D	E	F	G
Designed pH	6.8	7.2	7.6	8.0	8.4	8.8	9.2
Measured pH experiment I	6.85 $\pm$ 0.07	7.22 $\pm$ 0.07	7.62 $\pm$ 0.04	8.01 $\pm$ 0.02	8.37 $\pm$ 0.03	8.77 $\pm$ 0.02	9.18 $\pm$ 0.03
Measured pH experiment II	6.92 $\pm$ 0.15	7.28 $\pm$ 0.11	7.59 $\pm$ 0.01	8.03 $\pm$ 0.03	8.38 $\pm$ 0.03	8.78 $\pm$ 0.01	9.17 $\pm$ 0.02

Hatching rate

Figure 1 shows the hatching rates of geoduck clams at different pH values. Hatching rates were high at pH 8.0 ($94 \pm 1.3\%$), pH 8.4 ($91.11 \pm 1.4\%$), and pH 8.8 ($93 \pm 1.6\%$) but were lower at pH 6.8 and 9.2. The hatching rates of the embryos in the pH range 7.6 to 8.8 were significantly higher than those of the other groups ($p < 0.05$).

Larval survival and metamorphosis

Figures 2, 3, and 4 show larval survival, growth, and metamorphosis of geoduck clams. Larval survival rates of D stage larvae to day 7 were higher in the pH range 8.0 to 9.2 but lower at pH below 7.6 (Fig. 2). Growth was rapid at pH 8.4 and 8.8 but slow at pH 6.8 and 7.2 (Fig. 3). Differences in mean shell length were not significant among geoduck clams reared at pH 8.4 and 8.8, but the values were significantly lower than those of the other groups. Larvae were able to undergo metamorphosis when cultured in the pH range 7.6 to 8.8, but larvae cultured at lower and higher pH values did not reach metamorphosis and exhibited complete mortality at pH 9.2 (Fig. 4).

Juvenile survival and burrowing rate

The survival rate was significantly higher in juveniles reared at pH 8.0 and 8.4 than in the other pH conditions ($p < 0.05$), and survival rate did not differ significantly between pH 8.0 and 8.4. The larvae showed poor survival when the pH was below 7.6 or above 8.8 (Fig. 5).

The burrowing behavior of juvenile varied significantly among pH conditions (Fig. 6). After 24 h, juvenile clams burrowed into sediments at the highest rate (83.33%) at pH 8.4, and 58.33% of clams had burrowed into sediments at pH 8.8. One-way ANOVA revealed no

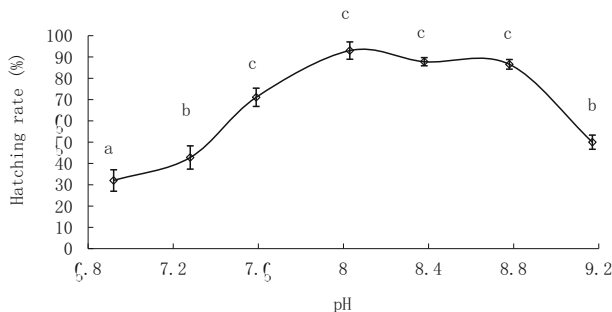


Fig. 1 Hatching rates of geoduck clams under different pH conditions. Different letters indicate hatching rate of geoduck differed significantly ($p < 0.05$). The same as follows

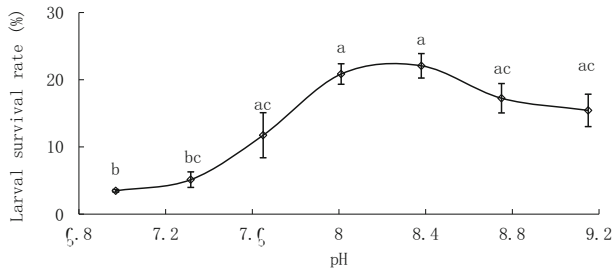


Fig. 2 Survival rates of geoduck clams under different pH conditions

significant effect of pH 7.2 to pH 8.0 on juvenile burrowing rates ($p < 0.05$). However, only 8.33% of juvenile clams burrowed into the sediment at pH 6.8.

Discussion

Our results show that hatching rate was significantly affected by pH. A gradual reduction of the hatching success was noted at pH values < 8.0 and > 8.8 . These findings demonstrate that pH influences the early stages of geoduck clams' life cycle, as it does in other clams (Parker et al. 2010). In addition, hatching rate at low pH (6.8) was lower than that at high pH (9.2), indicating that acidic pH was more lethal than the alkaline pH condition for geoduck clams. Hatching failure under such pH conditions has been reported for other species as well. For example, the mussel *Mytilus edulis* (Gazeau et al. 2010), oysters *Crassostrea virginica* (Miller et al. 2009), and the clam *Macoma balthica* (Van Colen et al. 2012) exhibited a low hatching rate at very high and low seawater pH in the culture environment.

Bivalve species are generally influenced by seawater pH. Dupont et al. (2010) reported that larval development is vulnerable to pH, and You et al. (1993) found that the highest survival rate of D-larvae of scallops occurred at pH 8.0. The D-shaped larval stage is an important stage in calcified organisms (Wang et al. 2016). The D-shaped larval stage can be used as an indicator for relative spawning periods, because it is the first larval stage after spawning (Brandner et al. 2017). A decrease in pH may negatively affect D-shaped larval shell formation, which can even stop in extreme conditions. Studies have reported are reduced

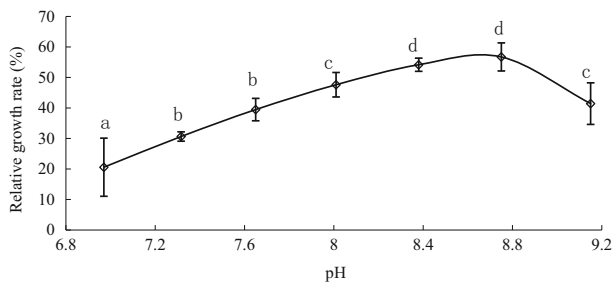


Fig. 3 Growth rates of geoduck clam larvae under different pH conditions

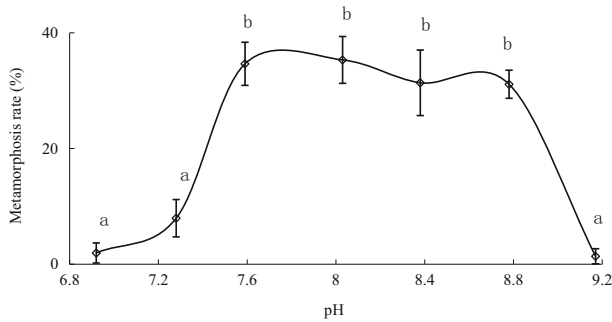


Fig. 4 Metamorphosis rates of geoduck clam larvae under different pH conditions

calcification rate in other marine organism and bivalves acidic pH (Jablonski and Lutz 1982; Kleypas et al. 2005). Wang et al. (2016) also observed a relatively smaller shell size of scallop due to the unusual formation of the shell. Reduction of calcification rate in acidic environment is responsible for this unusual shell formation. These findings suggest that changes in pH of geoduck clam culture water could have a lethal effect on D-shaped larvae as well as on juvenile and adult clams.

In this study, the survival rate and relative growth of larvae were calculated during the experimental period. The relative growth of larvae was significantly correlated with pH, which indicates that proper pH is essential for larval growth. Specifically, an alkaline environment was advantageous to geoduck clam growth. Calcification and mineralization are very important in the shell building process. An alkaline environment is favorable for calcification (Zhao et al. 2016), but calcification is impaired under acidic conditions. The main component of the shell is calcium carbonate. At alkaline pH values, sea water calcium carbonate is super saturated, which is conducive to the formation and stability of calcium carbonate, which in turn accelerates the growth and development of larvae. Several previous studies reported that alkaline conditions help marine bivalve growth (Waldbusser et al. 2014). Zhao et al. (2016) reported that alkaline conditions favored mussel growth through increased calcification and mineralization process. In the current study, the lowest growth rate was observed at pH 6.8, likely because lower pH hampers the mineralization process and results in reduced larval

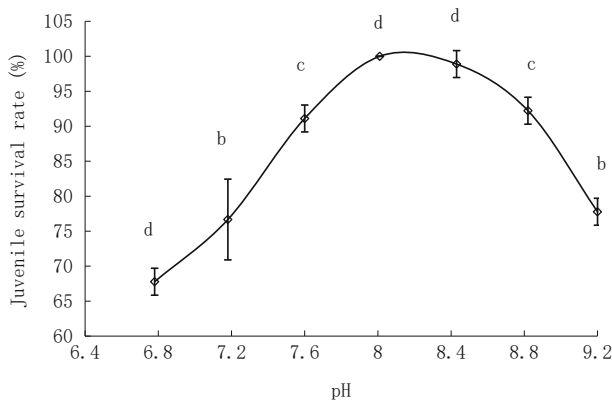


Fig. 5 Survival rates of juvenile geoduck clams under different pH conditions

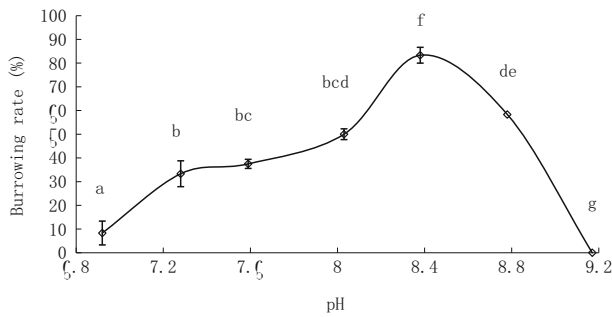


Fig. 6 Burrowing rates of juvenile geoduck clams under different pH conditions

growth. Studies of ocean acidification have reported that low pH impairs the shell calcification process and leads to reduced growth rate (Calabres and Davis 1996; Michaelidis et al. 2005; Gazeau et al. 2013; Milano et al. 2016), and, in extreme cases, mass mortality of the larval stage can occur (McClintock et al. 2009; Gazeau et al. 2013). However, the decreased survival rate of geoduck larvae at acidic pH might have resulted from physiological stress associated with metabolic failure. Metabolic suppression is, in many cases, an adaptive strategy for survival under transiently stressful conditions. Low pH levels trigger metabolic depression in many organisms (Hochachka and Guppy 1987; Guppy and Withers 1999; Abele et al. 2009). In shellfish, pH generally affects metabolism through reduction of protein synthesis (Kurihara et al. 2004; Kurihara 2008), which becomes more apparent in later stages of development. As a consequence, reduction of shell length and thickness has the potential to increase mortality rates during the larval stage. Similar results due to acidic pH were observed in the Pacific oyster (*Crassostrea gigas*) and the bay scallop *Argopecten irradians* (Talmage and Gobler 2009; Meseck et al. 2016).

pH had a significant effect on both metamorphosis and juvenile survival in this study. A significant difference in metamorphosis rate was observed in acidic and alkaline conditions outside the suitable range of pH 7.6 to 8.8. Highly acidic and highly alkaline pH delayed the geoduck metamorphosis process, and it almost stopped at pH 6.8 and 9.2. This suggests that pre-metamorphic larvae are very sensitive to pH. Prior to the short settlement and metamorphosis, deposited CaCO_3 in larval stage partially convert to aragonite. CaCO_3 might become more soluble at acidic pH and thereby hinder metamorphosis, which in turn might increase mortality rate of juvenile shellfish. In coastal areas, bivalves naturally experience extremely high mortality rates during the transition from larvae to benthic juveniles (Green et al. 2004). Tamburri and Zimmer-Faust (1996) reported that small individuals and those with extended metamorphosis would be susceptible to greater rates of predation and natural mortality during juvenile stages. Furthermore, a smaller size can reduce the feeding efficiency of larvae (Talmage and Gobler 2010), and smaller larvae are more susceptible to starvation because they encounter comparatively less food (Anger 1987; Hart and Strathmann 1994). The energetics of larval dispersal and altered energetics also may influence metamorphosis, as was reported for bivalves with impaired metamorphosis (Talmage and Gobler 2010). In studies of the Pacific oyster and the Olympia oyster (*Ostrea lurida*), a decrease in metamorphosis with decreased pH was reported (Hettinger et al. 2013; Gobler et al. 2014; Ko et al. 2014; Talmage and Gobler 2011). However, why a low level of metamorphosis at high pH was observed in geoduck clams in this study is unclear.

Burrowing behavior is a special feature of geoduck clams. A geoduck clam can bury itself more than 1 m deep in the gravel or sand of the ocean floor. Digging behavior is the direct result of foot muscular movement, and it is closely linked to the biological energy supply of the muscle (Gosling 2008). Geoduck clams have this ability during a certain period of their life cycle, but thereafter they lose digging ability. The shells of geoduck clams remain sedentary deep within the sand, with only the neck or siphon extending to the surface of the seafloor to feed on microscopic creatures. They can retract the siphon to protect themselves from predators and harsh environments. In this study, the best burrowing activity of juveniles occurred at pH 8.4, but altered digging behavior was observed at all experimental pH values relative to the control. Peng et al. (2017) reported a similar result for the razor clam *Sinonovacula constricta*. This result may be due to a series of interdependent and coordinated physiological changes induced by pH. Ion concentration changes could also disrupt neural signal transmission and subsequently affect the movement of the foot muscle.

In conclusion, the results of this study of geoduck larval growth and development demonstrate a tolerance range of pH values throughout the larval stages as well as the negative effects of variable pH at those stages. A suitable range of pH promotes faster growth and development in geoduck clams, but physiological abnormalities become apparent at pH values outside of this range. The adaptive physiological tolerance of larvae is not sufficient to withstand changing seawater pH. Further research is needed to understand the different biological processes affected by pH.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with animals performed by any of the authors.

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